



CS 551: Human Computer Interaction

Course Code	CS 551
Course Name	Human Computer Interaction
L-T-P-C	3-0-0-3
Prerequisites	A programming course at the undergraduate level involving the use of Python, C#, HTML, or JavaScript for interface design.
Intended for	B. Tech./Masters/Ph.D.
Mutual Exclusion	None
Approval	44th BoA (Nov 2021)

Course Contents

- **Introduction to HCI:** Course overview; goals in HCI; importance of design for usability; usability goals and metrics; historical perspective: machinery, computers, PCs and GUIs, and the Internet; different types of users; usability guidelines, principles, and theories of attention, perception, memory, and decision making. [8 Lectures]
- **User experience and design:** Different methods and frameworks of design; tools, practices, and patterns of design; social impact analysis; task decomposition; cognitive walkthroughs; expert reviews and heuristics; heuristic evaluation; guidelines of usability; active use evaluation; motion and time studies; GOMS keystroke-level models; human-study methodologies and techniques; survey and interview instruments; metaphors; storyboards; acceptance tests; ethical issues; design cases. [12 Lectures]
- **Interaction Design:** Direct manipulation; 2D devices and 3D interfaces; teleoperation and presence; augmented and virtual augmented reality; certain design patterns; fluid navigation; speech recognition and production; human language technology; traditional command languages; models of collaboration and contexts; deciding the appropriate interaction design. [10 Lectures]
- **Design choices:** Choices for user experience (animation, color, error handling, etc.); timing of user experience (system response time influence); help menus; information search; data types and data visualizations; grand challenges and future interfaces. [12 Lectures]

Text Books

1. Ben Shneiderman, Catherine Plaisant, Maxine Cohen, Steven Jacobs, Niklas Elmqvist, and Nicholas Diakopoulos, **Designing the User Interface: Strategies for Effective Human- Computer Interaction**, 6th Edition, Pearson, 2016
2. Dan R. Olsen Jr., **Building Interactive Systems: Principles for Human-Computer Interaction**, Cengage, 2010.

References

1. Jeff Johnson, **Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines**, 3rd edition, Morgan Kaufmann, 2020.
2. Jenifer Tidwell, Charles Brewer, Aynne Valencia, **Designing Interfaces: Patterns for Effective Interaction Design**, 3rd Edition, O'Reilly Media, 2020
3. Materials from other online sources [<http://link1>], [<https://www.inf.ed.ac.uk/teaching/courses/hci/1718/>], and [<http://pages.cpsc.ucalgary.ca/~saul/481/index.html>]